

# Barash Sharma

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## EDUCATION

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| <b>School of Engineering, Cochin University of Science and Technology (CUSAT)</b><br><i>Bachelor of Technology (B.Tech) in Computer Science and Engineering, CGPA: 8.5/10</i> | Cochin, Kerala<br>Oct 2022 – Apr 2026 |
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## TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, SQL
- **Web & Backend:** Spring Boot, Node.js, FastAPI, REST APIs, JWT, Microservices
- **Frontend & Mobile:** React, Next.js, Tailwind CSS, React Native
- **Databases & DevOps:** PostgreSQL, MongoDB, Redis, Docker, AWS (EC2, S3, RDS), CI/CD
- **ML & Tools:** PyTorch, CNNs, Transformers, Firebase

## EXPERIENCE

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| <b>Ylogx Pvt. Ltd.</b><br><i>Software Engineer Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Feb 2025 – Dec 2025 |
| <ul style="list-style-type: none"><li>• Modularized admin dashboards in <b>React and Node.js</b> enabling role approvals, access management, and workflow control of about <b>50 employees</b>, similar to AWS-style permission handling.</li><li>• Built <b>Express.js middleware</b> for <b>JWT authentication</b> and <b>Role-Based Access Control (RBAC)</b>, automating permission checks that were previously handled manually via spreadsheets.</li><li>• Constructed a <b>CI/CD pipeline</b> using <b>GitHub Actions</b> to automate multi-stage deployments to <b>AWS EC2, S3, and RDS</b>, incorporating <b>SQL migration scripts</b> for database consistency.</li><li>• Reduced average page load time by <b>25%</b> by using <b>React lazy</b> for code-splitting and optimizing <b>SQL join queries</b> to reduce API response sizes.</li></ul> |                     |
| <b>IIT Jammu</b><br><i>Application Developer Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | May 2023 – Sep 2024 |
| <ul style="list-style-type: none"><li>• Developed a campus chatbot using <b>React Native and FastAPI</b> that resolved <b>1,500+ queries/month</b> by implementing a <b>conditional routing engine</b> for student FAQ data.</li><li>• Architected a <b>PostgreSQL</b> schema with <b>indexing on high-frequency columns</b> and integrated <b>AsyncStorage</b> for local caching to maintain app functionality during network drops.</li><li>• Created a <b>Web Admin Portal</b> that allows staff to modify chatbot responses and user roles via <b>RESTful APIs</b>, eliminating direct manual edits to the database.</li></ul>                                                                                                                                                                                                                            |                     |

## PROJECTS

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| <b>Single Image Super-Resolution (Ongoing)</b>                                                                                                                                                                                                                                                                                                                                                                                                                             | Jan 2026 – Present  |
| <ul style="list-style-type: none"><li>• Designing a hybrid <b>CNN and Transformer</b> model in <b>PyTorch</b> for single-image super-resolution, aiming to enhance image clarity and PSNR metrics.</li></ul>                                                                                                                                                                                                                                                               |                     |
| <b>Geospatial Flood Analysis System (PostGIS)</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | Jan 2026 - Feb 2026 |
| <ul style="list-style-type: none"><li>• Designed and implemented a <b>PostgreSQL + PostGIS</b> geospatial database using India-wide OpenStreetMap (OSM) data, enabling spatial queries on roads, rivers, buildings, and administrative boundaries.</li><li>• Built a <b>Natural Language to Spatial SQL pipeline</b> that generates validated PostGIS queries with GIST indexing, region-based filtering, and GeoJSON/Shapefile exports for flood-risk analysis.</li></ul> |                     |
| <b>VertEase – Vertigo Management App</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   | Jan 2025 – Mar 2025 |
| <ul style="list-style-type: none"><li>• Built a mobile health tracker using <b>React Native</b> for symptom logging and <b>FastAPI</b> for data persistence.</li><li>• Containerized the FastAPI backend and React frontend using <b>Docker</b> for deployment on <b>AWS EC2</b>, while integrating <b>Firebase</b> for real-time data synchronization.</li></ul>                                                                                                          |                     |